

A Deep Learning-based Web Application for CT reconstruction from X-ray image

Huu Quang Hoa Nguyen^{1†} and Tran Quoc Bao Le^{1†} and Quy Dat Tran^{1†}

Thi Tan Tien Nguyen^{2 3*}[0000-0002-6300-336X] and Hang Phuong Nguyen⁴[0000-0003-0801-9702]

Cong Tran¹[0000-0001-9467-4978] and Cuong Pham¹[0000-0003-0973-0889]

¹ Posts and Telecommunications Institute of Technology, Hanoi, Vietnam

² Thai Nguyen University of Information and Communication Technology, Thai Nguyen, Vietnam

³ Thai Nguyen University of Medicine and Pharmacy, Thai Nguyen, Vietnam

⁴ School of Mechanical Engineering, University of Ulsan, Ulsan, Republic of Korea

*Corresponding author(s). E-mail(s): nguyenthitantien@tnmc.edu.vn

[†]These authors contributed equally to this work.

Abstract. This study proposes a web application to address the challenge of accessibility to Computed Tomography (CT) equipment by reconstructing 3D spatial information from a 2D X-ray image. Our proposed method is a multi-stage deep learning pipeline including generating synthetic data from DiffDRR, reducing the domain gap between synthetic and real images by CycleGAN, and reconstructing the CT image from X-ray2CTPA architecture. The results present that the proposed framework not only reconstructs 3D volumes with consistent anatomical structures but also allows users to interact with rotation and viewing slices (coronal, sagittal, axial) via the web interface. This study hopes to contribute to the collective effort of applying AI to enhance the informational value of common diagnostic imaging modalities, thereby opening up the potential for improved healthcare accessibility for the community.

Keywords: Computed tomography reconstruction, CycleGAN, DiffDRR, Synthetic-to-Real Adaptation.

1 Introduction

Computed Tomography (CT) plays a critical role in computer-assisted surgery due to its ability to provide high-contrast cross-sectional imaging of internal structures [1]. However, acquiring volumetric CT images requires multiple X-ray projections from different angles, leading to long scan times and significantly higher radiation exposure than standard X-rays [2], [3]. Furthermore, CT equipment is expensive and not always available in under-resourced regions or lower-level medical facilities [4].

In contrast, 2D X-ray imaging is widely accessible and commonly used in clinical practice, however, its limitation lies in the loss of depth information, especially when anatomical structures overlap. This makes it challenging to understand spatial relationships between lesions. Therefore, developing an efficient method to reconstruct 3D CT-like images from safer and cheaper 2D X-rays is crucial.

Early deep learning-based approaches primarily utilized CNNs, U-Net [5], and GAN variants to directly map 2D X-rays to 3D CT volumes [6], [7]. Although the frameworks are relatively simple, these methods require large-scale datasets and precisely aligned X-ray/CT pairs, which are both costly and challenging to collect. To address these challenges, many studies shared the ideas of generating synthetic X-rays (Digitally Reconstructed Radiographs – DRRs) from publicly available CT datasets [8]. This strategy reduces the limitation of acquiring real corresponding paired data, but still introduces a domain gap between clean synthetic X-ray images and noisy clinical X-rays. More recent studies have proposed advanced solutions to bridge this gap, such as Dual-View Guided Diffusion Model (DVG-Diffusion) [9], Diffusion Learning to Reconstruct CT Images (DiffuX2CT) [10], and Large Gaussian Reconstruction Model for Sparse-view X-rays (X-GRM) [11]. The DVG-Diffusion [9] reconstructs the CT images by (1) synthesizing an additional projection from a single-projection reality X-ray and (2) leveraging both projections to guide a latent diffusion model for 3D reconstruction. The DiffuX2CT [10], in another way, adopts a conditional diffusion framework that is trained on reality-paired biplanar X-rays and lumbar spine CT to enable mapping directly from multiple X-ray views to volumetric CT. Recently, the study of X-GRM [11] has generated synthetic X-rays from large-scale public CT datasets using Tomographic Iterative GPU-based Reconstruction (TIGRE) Toolbox [12], by injecting realistic noise to mimic clinical conditions, and then applying a Gaussian representation with Transformer-based modeling to recover CT volumes. Although these approaches present potential results, their performances are still limited by the accumulated errors in synthesized projections as well as large-scale design. Furthermore, the domain gap between the clean synthetic X-ray images and noisy clinical X-rays has been challenging.

To overcome these issues, we propose an application-driven, web-based framework that reconstructs 3D CT volumes from single 2D X-rays and enables interactive visualization. The system includes: (1) synthetic X-ray generation via DiffDRR [13], (2) domain adaptation using CycleGAN [14], and (3) 3D reconstruction through the X-ray2CTPA architecture [15]. The model is trained and validated using the public LIDC-IDRI dataset [8], aiming to reduce cost and increase accessibility to 3D imaging, particularly for resource-limited settings.

2 Methodology

Our approach combines a multi-stage deep learning pipeline with a patient-centric application framework, as shown in **Fig. 1**. The core of this approach is a multi-stage deep learning pipeline meticulously designed to address the inherent challenges of paired data scarcity and the domain discrepancy between synthetic and real-world clinical data, encapsulated within a patient-centric application framework, aiming to democratize access to and understanding of medical imaging data. To tackle the lack of paired medical images, the CycleGAN [14] was employed in this study. The principle of CycleGAN [14] enables synthetic X-rays to approximate real clinical X-rays without requiring exact correspondence. The cycle consistency loss of CycleGAN [14] also plays a key role in maintaining essential anatomical structures in the synthesized X-rays.

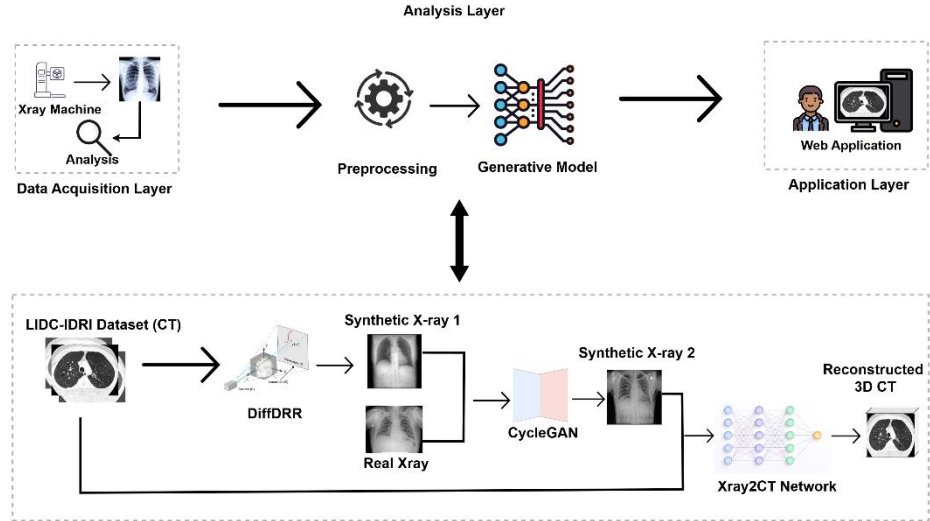


Fig. 1. A detailed architectural diagram of the system. The upper part describes the user-centric workflow. The lower part presents the core technical pipeline, consisting of three main steps: (1) Generating synthetic X-rays from CTs using DiffDRR [13], (2) Adapting the image domain using CycleGAN [14], and (3) Reconstructing the 3D CT from the X-ray using the X-ray2CTPA [15] architecture.

We developed a web-based application designed for direct patient use, transforming an abstract 2D X-ray into an interactive 3D model. It consists of three layers: a Data Acquisition Layer with a user-friendly interface for image uploads; an Analysis Layer that runs the core deep learning pipeline to reconstruct the 3D volume; and an Application Layer for interactive visualization, empowering patients with a clearer understanding of their health status.

2.1 Datasets and preprocessing

We utilized the publicly available Lung Image Database Consortium (LIDC-IDRI) dataset, which contains 1012 chest CT scans [8]. All CT volumes were resampled to an isotropic voxel spacing of $1 \times 1 \times 1$ mm, and intensity values for all images were normalized to the range $[-1, 1]$ for training stability.

2.2 Synthetic X-ray Generation via Differentiable Rendering

To overcome the lack of paired X-ray/CT data, we generate synthetic X-rays (DRRs) from CT volumes using DiffDRR [13], a differentiable renderer. This process simulates the Beer-Lambert law, where the DRR pixel value is proportional to the line integral of attenuation coefficients $\mu(x)$ along a ray path R :

$$\int_R \mu(x) dx \approx \sum_k l_k \cdot V_k \quad [1] \quad (1)$$

where V_k is the voxel value and l_k is the path length of the ray through voxel k . DiffDRR [13] reframes this process using vectorized tensor operations, making it fully differentiable and GPU-accelerated. This yields a dataset of geometrically faithful synthetic X-rays (termed *Synthetic X-ray 1*).

2.3 Unsupervised Domain Adaptation with CycleGAN

A significant domain gap exists between clean *Synthetic X-ray 1* images and noisy clinical X-rays. To bridge this, we employ CycleGAN [14], based on the principles of Generative Adversarial Networks (GANs) [16], an unsupervised image-to-image translation framework. CycleGAN [14] learns two mappings, $G: X \rightarrow Y$ (synthetic-to-real) and $F: Y \rightarrow X$ (real-to-synthetic), using a combination of an adversarial loss and a cycle consistency loss. The cycle consistency loss ensures structural content is preserved by enforcing that $F(G(x)) \approx x$ and $G(F(y)) \approx y$. The loss is defined as:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{\text{data}}(y)} [\|G(F(y)) - y\|_1] \quad [14] \quad (2)$$

The full objective function of CycleGAN [14] is a weighted sum of the two components:

$$\mathcal{L}(G, F, D_X, D_Y) = \mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) + \mathcal{L}_{\text{GAN}}(F, D_X, Y, X) + \lambda \mathcal{L}_{\text{cyc}}(G, F) \quad [14] \quad (3)$$

where λ is a hyperparameter that controls the relative importance of the cycle consistency loss. The output of this stage is images termed *Synthetic X-ray 2*, a hybrid dataset that possesses the anatomical structure of CTs and the image characteristics of clinical X-rays, ready for the final 3D reconstruction step.

2.4 2D-to-3D Reconstruction using X-ray2CTPA Architecture

The final stage reconstructs the 3D CT volume using the X-ray2CTPA architecture [15], a state-of-the-art conditional diffusion model. Based on DDPMs [17], it uses a 3D U-Net [5] to reverse a Gaussian noise process, conditioned on a feature vector c extracted from the input X-ray by a pre-trained encoder like BiomedCLIP [18]. The model is optimized by minimizing the loss:

$$\mathcal{L}_{\text{DM}} = \mathbb{E}_{t, z_0, \epsilon, c} [\|\epsilon - \epsilon_\theta(z_t, c, t)\|^2] \quad [15] \quad (4)$$

where ϵ_θ is the U-Net [5] and z_t is the noised CT data at timestep t . The model is trained on pairs of *Synthetic X-ray* 2 images and their corresponding ground truth 3D CT volumes [8].

3 Experiments

3.1 Data Preparation

For this study, the publicly available LIDC-IDRI lung CT dataset [8] was utilized to generate a set of synthetic X-ray images via Differentiable Digitally Reconstructed Radiograph (DiffDRR) technology [13]. The CycleGAN [14] model was trained in 1000 epochs with training parameters including input image size of 256x256, the Adam optimizer, and a learning rate of 0.0002. The final training dataset for CT 3D Reconstruction consisted of paired images, each including a domain-adapted X-ray and its corresponding original 3D CT volume from the LIDC-IDRI datasets [8].

3.2 Implementation and Training Details

To address the 2D X-ray to 3D CT conversion challenge, this study proposes a Conditional Diffusion Model [19] optimized for web-based applications.

The core architecture is a latent diffusion model. The generation process begins with a Gaussian noise tensor of size (4, 64, 32, 32), conditioned by a 512-dimensional feature vector extracted from an X-ray image of size (224, 224, 3) using the BiomedCLIP [18] Visual Encoder. A U-Net 3D network [5] with 4 symmetric levels ($\text{dim_mults}=[1,2,4,8]$) and an 8-head Attention mechanism performs the denoising process over 1000 DDPM [17] sampling steps. The conditioning vector is integrated via adaptive modulation within the Resnet Blocks [20]. Finally, the decoder of a 2D Auto-encoderKL model reconstructs the clean latent tensor of size (4, 64, 32, 32) into 64 image slices of size (1, 256, 256), which form the complete 3D CT volume.

The model is trained for 1,000,000 steps with a batch size of 10, using the AdamW optimizer and a Cosine Annealing scheduler. The learning rate is set to $1\text{e-}4$ for pre-training and $1\text{e-}5$ for fine-tuning. The model was trained using a composite loss function that incorporates both an L1 reconstruction loss and a perceptual loss. This ensures

that the generated CT volumes are not only structurally accurate but also preserve critical clinical details. For performance evaluation, 10% of the LIDC-IDRI dataset [8] was held out as a test set, on which the reconstruction quality was measured using the Structural Similarity Index (SSIM) [21] and Peak Signal-to-Noise Ratio (PSNR) [22].

4 Results

4.1 Web Application for Patient Visualization

To translate the research findings into a practical tool, this study developed a web application for patient-oriented visualization, allowing users to upload their X-ray images and receive a corresponding 3D CT scan reconstructed by the model.

The application's interface is designed to be intuitive, empowering users to interact with and comprehend their own medical imaging data through a comprehensive suite of features. Key functionalities include displaying images across standard anatomical planes (Axial, Sagittal, and Coronal) with a navigation slider, and a simplified windowing adjustment tool to highlight specific tissue types.

By integrating these features, the application serves not only as a tool to demonstrate the study's results but also as a supportive platform to enhance the patient's experience and understanding.

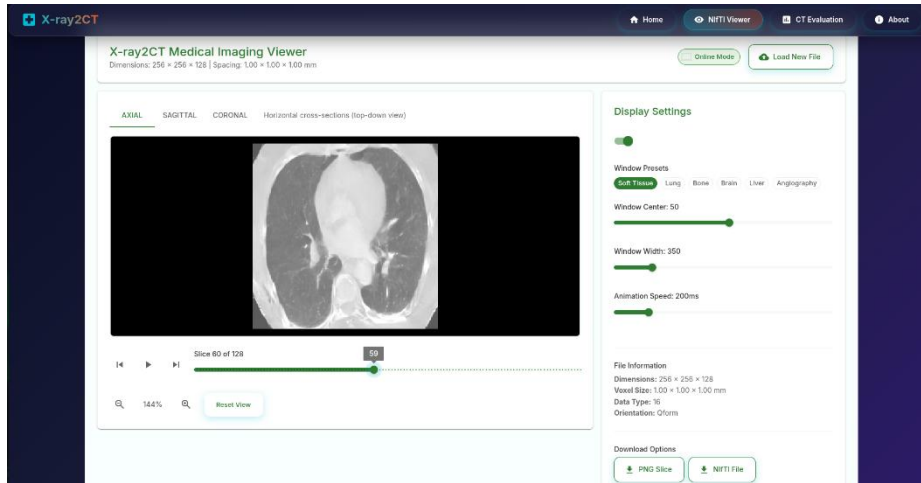


Fig. 2. The main interface of the X-ray2CTPA web application.

4.2 Quantitative Evaluation

To objectively assess the model's performance, a quantitative evaluation was conducted on a hold-out test set, which comprised 10% of the LIDC-IDRI datasets [8] reserved

specifically for this purpose. Two prevalent and critical metrics for image quality assessment, the Structural Similarity Index (SSIM) [21] and the Peak Signal-to-Noise Ratio (PSNR) [22], were utilized.

Table 1. Quantitative evaluation of image quality

SSIM	PSNR
0.7017	24.79 dB

The results presented in Table 1 indicate that the model achieved a Structural Similarity Index (SSIM) [21] of 0.7017 and a Peak Signal-to-Noise Ratio (PSNR) [22] of 24.79 dB. These values, considered to be relatively favorable, suggest that the reconstructed CT images exhibit good structural similarity and preserve the overall characteristics of the ground truth images.

A detailed comparison reveals that the "Reconstructed CT" exhibits a high degree of fidelity to the "CT Ground Truth" in major structures. The morphology of the lungs, the cardiac silhouette, and the primary bronchi are accurately reproduced in terms of position, shape, and relative scale. This structural alignment serves as a significant validation, suggesting that the model has learned to infer complex three-dimensional spatial relationships from two-dimensional input data.

4.3 Qualitative Evaluation

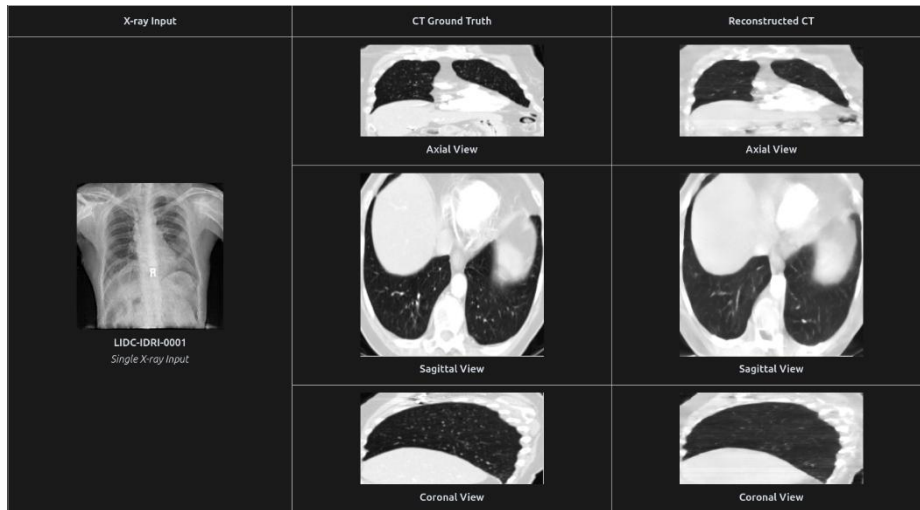


Fig. 3. Visual assessment of the CT Reconstruction from X-ray image.

5 Discussion

This study proposed an end-to-end web-based system for reconstructing and interacting with 3D CT images from 2D X-rays. The framework effectively addresses two major challenges: the lack of paired data and the domain gap between synthetic and real clinical X-rays, while also offering an intuitive visualization experience, especially for patients.

As shown in **Table 1**, the model achieved a Structural Similarity Index (SSIM) [21] of 0.7017 and a Peak Signal-to-Noise Ratio (PSNR) [22] of 24.79 dB, indicating that the reconstructed CT images preserve overall anatomical structure with acceptable image quality. In addition, **Figure 3** clearly demonstrates that the model successfully reconstructs the position and shape of key anatomical features such as the lungs, heart, and main bronchi.

However, fine anatomical details, such as peripheral pulmonary vessels, are still less sharp compared to the original CT scans. This limitation is common in cross-modality image synthesis and represents a direction for future improvement. Nevertheless, the system has shown its capability to generate anatomically plausible CT volumes, fulfilling its primary goal of enhancing patient understanding and engagement through interactive 3D visualization.

6 Conclusion

This paper proposes an application framework to reconstruct 3D CT from a two-dimensional X-ray, with the goal of contributing to bridging the gap in access to advanced diagnostic imaging. Our method is built upon a multi-stage pipeline that combines dif-fDRR [13], CycleGAN [14], and X-ray2CTPA [15], designed to simultaneously handle the challenges of data scarcity and domain difference. This technical foundation enables an interactive web application that transforms 2D input data into an intuitive 3D model, not only supporting specialists but also helping patients to more easily visualize their own condition. Although still in its early stages and requiring further validation, we believe this work is a meaningful step, contributing to the collective effort of practically applying AI to enhance the value of common diagnostic methods.

Our future work will focus on enhancing the system's intelligence, moving beyond mere reconstruction. An ambitious direction is the integration of an Interactive Medical Chatbot, utilizing multi-modal learning to simultaneously analyze both the original 2D X-ray and the reconstructed 3D CT volume. The objective is for the chatbot to provide preliminary analyses or suggested diagnoses, transforming the system from an image visualization tool into an intelligent screening support assistant.

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